I Definitions: build context

II Proofs: use established context

a Introduce a real object

The declaration itself is checked

`have x R`

x is a real number

Check the carrier R
R is well-defined

Check that R is nonempty
StandardSetNonempty builtin

Register x and its declaration
record the membership $x \in \mathbb{R}$

Current visible scope Γ

$x \in \mathbb{R}$
`from: have x R`
x is a real number

declaration · membership · source

Declaration checks can invoke proof rules.

b Search for a proof

Check a new fact in the established context

$0 \le (x + 1)^2$
Classify the fact

Check well-definedness
objects · signature · domains

Truth search: try allowed routes in order

- Direct evidence
- Known facts / special properties
- Builtin rule
- Strategy
- Definition
- Known forall
- Rewrite (final fallback)

No proof found
all allowed routes miss

Return proof evidence

Premises inherit restricted search permissions.

c Composed proof evidence

The selected rule and its checked premise

$0 \le (x + 1)^2$

Even real powers are nonnegative

$x + 1 \in \mathbb{R}$

Real addition is closed

$x \in \mathbb{R}$ | $1 \in \mathbb{R}$

Context citation | Exact calculation

AND: both premises are required

Builtin rule → Direct → context / calculation

Updated context ← **Direct inference, once** ← **Store** ← **Verified fact**

required condition · returned evidence · mathematical rule